Subject: -	Report on fluid economy evaluation of TML Signa 4221.T Cx 5LNGE (204 HP engine) OBD-II BS-VI vehicle with G1150 (FGR: 9.13) Transmission and 5.85 Rear axle ratio in comparison with TML Signa 4221.T BS-VI OBD-I vehicle as a Base vehicle.

1.0 Conclusion:

1.1 Fluid economy in rated load condition at 40 kmph :

- Fluid economy of Signa 4221.T Cx BS-VI OBD-II vehicle was found **3.16 kmpl** which is **at par** wrt Signa 4221.T BS-VI OBD-I Base vehicle.

1.2 Fluid economy in rated load condition at 55 kmph :

- Fluid economy of Signa 4221.T Cx BS-VI OBD-II vehicle found **3.00 kmpl** which is **at par** wrt Signa 4221.T BS-VI OBD-I Base vehicle.

Action by: -- Failure class: --

1.3 Fluid economy in Unladen condition at 40 kmph :

- Fluid economy of Signa 4221.T Cx BS-VI OBD-II vehicle found **6.36 kmpl** which is **Superior by 2.91%** wrt Signa 4221.T BS-VI OBD-I Base vehicle.

Action by: -- Failure class: --

1.4 Fluid economy in Unladen condition at 55 kmph :

- Fluid economy of Signa 4221.T Cx BS-VI OBD-II vehicle found **5.96 kmpl** which was which was **Superior by 4.20%** wrt Signa 4221.T BS-VI OBD-I Base vehicle.

Action by: -- Failure class: --

1.5 Overall fluid economy status at 40 kmph (80% Laden + 20% unladen condition):

- Fluid economy of Signa 4221.T Cx BS-VI OBD-II vehicle found **3.52 kmpl** which is **at par** wrt Signa 4221.T BS-VI OBD-I Base vehicle.
- Signa 4221.T Cx BS-VI OBD-II meets the PALS target of “Leadership”.

1.6 Overall fluid economy status at 55 kmph (80% Laden + 20% unladen condition):

- Fluid economy of Signa 4221.T Cx BS-VI OBD-II vehicle found **3.33 kmpl** which is **at par** wrt Signa 4221.T BS-VI OBD-I Base vehicle.
- Signa 4221.T Cx BS-VI OBD-II meets the PALS target of “Leadership”.

Note: Concluded fluid economy results is extrapolated for Cx variant to reducing the THU bearing contribution by 2.5% from actual fluid economy results.

Action by: -- Failure class: --

2.0 Result: Vide Observations

FAILURES CLASS:

A: Failure that happen without warning. Detrimental to vehicle / personnel, road users and safety. Can cause death / hospitalization. Noncompliance of regulatory requirement.

B: Vehicle gets crippled with sufficient warning. Minor injury to occupant & road users (First aid only). Premature failures / repeated failures. Leads to consequential catastrophic damages. Need immediate attention.

C: Customer irritants. Capable of damaging aesthetics. Failure can be attended at next servicing Schedule.

D: Customer perception issues and minor irritants.

3.0 Test Vehicles: -

1. Signa 4221.T BS-VI OBD-II (ERCJ-1253)_Test vehicle

   Chassis no. - PC03509022RJ00800; Engine no. - GXX103502.

2. Signa 4221.T BS-VI OBD-I (ERCJ 1153)_Base vehicle

   Chassis no. - PC13518022RJ00784; Engine no. - GXX103780.

Enclosures:

Annexure 1: Panel chart

Annexure 2: PAT sheet

Annexure 3: Duty cycle report

Annexure 4: Test request

Annexure 5: Test plan

Input Doc Ref.: - Test Request No: - Testing-Outdoor/15689

DID No: – NA.

DDP No: – NA.

Work Order No. :- P.22.EA.458	**Make :-** TATA Motors Limited	**T I No. – 42355**

Earlier Ref: - Nil.

4.0 Project Status:- DR3

5.0 Introduction:

As a part of RDE (OBD-II) Testing Program, TML Signa 4221.T Cx 5LNGE BS-VI OBD-II vehicle with G1150 (FGR:9.13) Transmission and 5.85 Rear axle ratio was offered for fluid economy evaluation along with TML Signa 4221.T 5LNGE BS-VI OBD-I vehicle as a base vehicle. All vehicles were tested back-to-back at JSR- Ranchi-JSR route in laden and unladen conditions at 40 km/hr as well as 55 km/hr vehicle cruising speeds. This report will reveal details of this trial with the status.

6.0 Testing:

Following test was conducted during evaluation :

Sr. no.	Test	Reference standard	Instruments used
1	Fluid economy evaluation	TST/VT/WG/158	Data Logger

7.0 Vehicle details:

Parameter		Base Vehicle	Test vehicle
Vehicle model		Signa 4221.T (OBD-I) (ERC-1153)	Signa 4221.T (OBD-II) (ERC-1253)
A) Vehicle specification			
Engine Model		Tata 5LNGE BS-VI (200 HP) OBD-I	Tata 5LNGE BS-VI (204 HP) OBD-II
Max Power		200 HP @2200 RPM	204 HP @2200 RPM
Max Torque		850 Nm @ 1000 ~1600 rpm	850 Nm @ 1000 ~1600 rpm
GVW (kg)		42000	42000
Gear Box		Model: Tata G1150, Type: DD	Model: Tata G1150, Type: DD
Gear Ratios.		1st gear:9.13, 2nd gear: 6.72, 3rd gear: 4.90, 4th gear: 3.57, 5th gear: 2.55, 6th gear: 1.88, 7th gear: 1.37, 8th gear:1.00	1st gear:9.13, 2nd gear: 6.72, 3rd gear: 4.90, 4th gear: 3.57, 5th gear: 2.55, 6th gear: 1.88, 7th gear: 1.37, 8th gear:1.00
Axle Ratio		5.85	5.85
Tires	FA1	Size: 295/90R20, Make: JK , Type: non LCRR	Size: : 295/90R20, Make: M/s J K, Type : LCRR (Phase-II)
	FA2	Size: 295/90R20, Make: JK , Type: non LCRR	Size: : 295/90R20, Make: M/s J K, Type : LCRR (Phase-II)
	LA1	Size: 295/90R20, Make: JK , Type: non LCRR	Size: : 295/90R20, Make: M/s J K, Type : LCRR (Phase-II)
	RA1	Size: 295/90R20, Make: JK , Type: non LCRR	Size: : 295/90R20, Make: M/s J K, Type : LCRR (Phase-II)
	RA2	Size: 295/90R20, Make: JK , Type: non LCRR	Size: : 295/90R20, Make: M/s J K, Type : LCRR (Phase-II)
Tractive effort @ 1st gear		87390 N	87390 N
Tractive effort @ Top gear		9572 N	9572 N
Fan Details (Type/Make/Blades)		Type: E-Viscous fan and Size:24"	Type: E-Viscous fan and Size:24"

Parameter	Base Vehicle	Test vehicle
Vehicle model	Signa 4221.T (OBD-I) (ERC-1152)	Signa 4221.T (OBD-II) (ERC-1253)
B) FE Avenues _Engine Level		
Base Cal Improvement (Cal Maturity %)	As per production released	95% Cal Maturity
Low Viscous Engine Oil (TMGO-10W30)	No (15W40CK4+)	Yes (5W30)
FE Switch	Yes (3 Mode : E/B/P)	Yes (3 Mode : E/B/P)
Engine Features (EOL : Enable/Disable)	NA	NA
ATS Optimization	No	No
APM	No	No
Cruise Control	No	No
Gear based VAM	NA	NA
Intelligent mode shifter	NA	NA
Moving Mass Estimation	NA	NA
C) FE Avenues _Vehicle Level		
Driveline Optimization	NA	NA
GSA (Gear Shift Advisor)	Yes	Yes (Not Tuned)
MSA (Mode Shift Advisor)	NA	Yes (Not Tuned)
Low CRR Tyres	Yes	Yes
Low Viscous rear axle oil (80W90)	Yes	Yes
Smart Alternator (LIN)	NA	NA
THU/Unitised/Low friction bearing	No	Yes

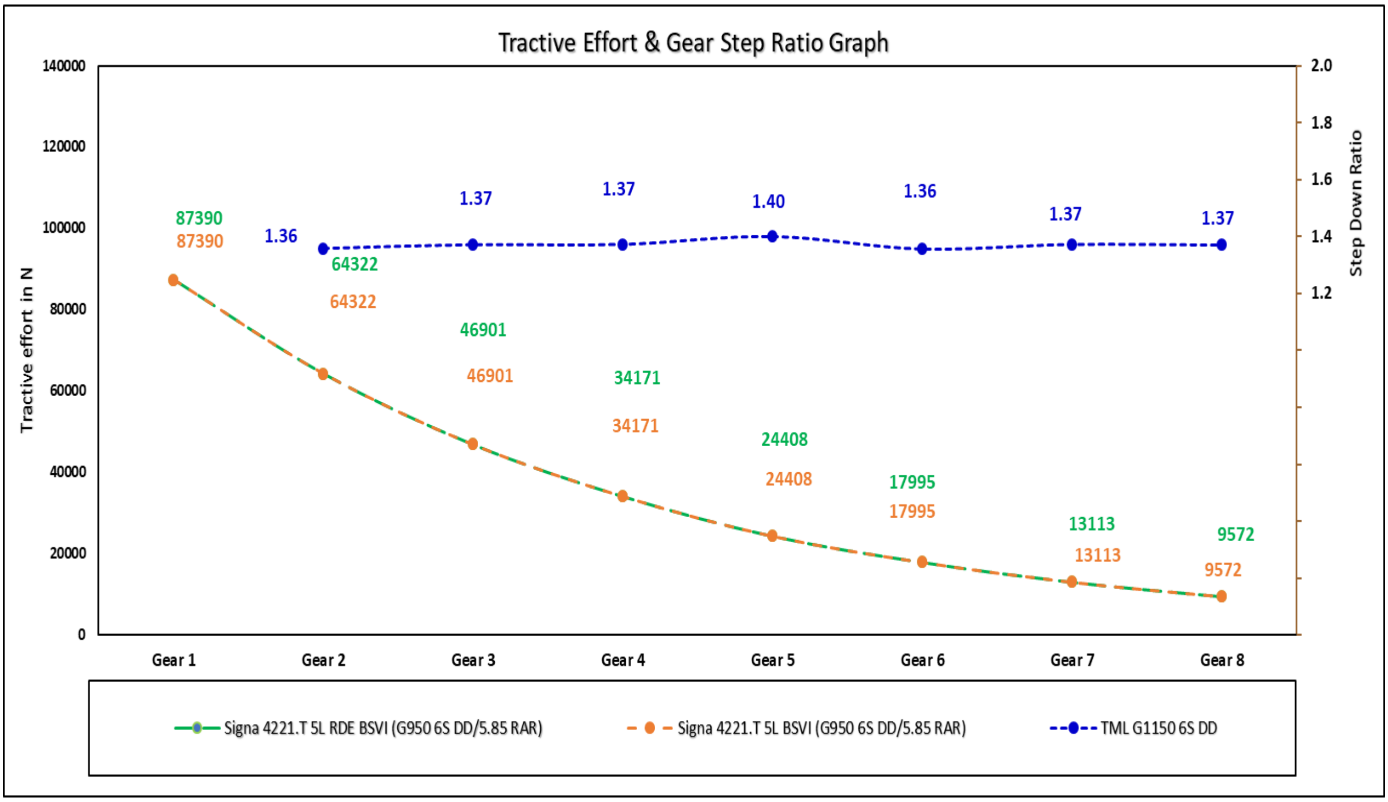
8.0 Test result:

8.1 Cruising speed: 40 Kmph & 55 Kmph :

Sr. No.	Parameters		Base Vehicle : OBD-1 SIGNA 4221.T Tata 5L BSVI								Test Vehicle : OBD-2 SIGNA 4221.T Tata 5L RDE BSVI							
1	Vehicle Identification No. (ERC No. /EJ No./Reg. No.)		ERC-1164								ERC-1259							
2	Vehicle GVW/GCW	kg	42000								42000							
3	Engine Details (Make/cc/Norms/no. of Cyl.)		TML 5LNGE								TML 5LNGE							
	Engine Max Power		200 HP @ 2200 rpm								204 HP @ 2200 rpm							
	Engine Max Torque		850 Nm @ 1000-1600 rpm								850 Nm @ 1000-1600 rpm							
	Engine Calibration Maturity Level		As per production released								95%							
4	Regeneration Strategy (Time ase/Soot base)		110 Hrs								110 Hrs							
5	Gear Box (make/drive/FGR)		**TATA G-1150 DD**								**TATA G-1150 DD**							
	Gear Ratios		1-9.13, 2- 6.72, 3- 4.90, 4- 3.57, 5- 2.55, 6- 1.88, 7- 1.37, 8- 1.00								1-9.13, 2- 6.72, 3- 4.90, 4- 3.57, 5- 2.55, 6- 1.88, 7- 1.37, 8- 1.00							
6	RAR		5.85								5.85							
7	Tyre Details		295/90 R 20 RIB JK Non-low CRR								295/90R 20 RIB JK Low CRR							
8	RSG — Maximum Accelerator Vehicle Speed	kmph	80								80							
	USFE / Torque Table Revision (1/2/3/n)		--								--							
9	Fan Details (type/make/blades)		E- 24" High Efficiency Viscous fan								E- 24" High Efficiency Viscous fan							
FE Avenues (Engine	Base Cal Improvement (Cal Maturity %)		As per production released								--							
	Low Viscous Engine Oil (TMGO-10W30)		No								Yes							
	FE Switch		Yes (3 Mode : E/B/P)								Yes (3 Mode : E/B/P)							
	ATS Optimization		As per production released								No							
	APM		As per production released								No							
FE Avenues (Vehicl	Driveline Optimization		NA								No							
	GSA (Gear Shift Advisor)		Yes								Yes (Not Tuned)							
	Low CRR Tyres		No								Yes							
	Low Viscous rear axle oil (80W90)		Yes								Yes							
	THU/Unitised/Low friction bearing		No								Yes							
FE Trial Details	Route		**Jamshedpur-Ranchi-Jamshedpur**								**Jamshedpur-Ranchi-Jamshedpur**							
	FE Duty Cycle (Laden/Unladen)		**Laden Trial @ 40 kmph**		**Unladen Trial @ 40 kmph**		**Laden Trial @ 55 kmph**		**Unladen Trial @ 55 kmph**		**Laden Trial @ 40 kmph**		**Unladen Trial @ 40 kmph**		**Laden Trial @ 55 kmph**		**Unladen Trial @ 55 kmph**	
	Trial Date		07.06.2023	08.06.2023	15.06.2023	16.06.2023	05.06.2023	06.06.2023	17.06.2023	19.06.2023	07.06.2023	08.06.2023	15.06.2023	16.06.2023	05.06.2023	06.06.2023	17.06.2023	19.06.2023
	Trip Distance	km	241	241	248	241	241	241	241	241	241	241	248	241	241	241	241	241
	Total Diesel Quantity	Litre	72.64	74.41	40.41	37.03	76.61	80.00	40.29	42.37	71.53	72.85	38.23	35.59	75.19	78.21	37.62	40.2
	Total Urea Quantity	Litre	4.50	4.60	1.70	1.90	3.40	3.90	1.60	1.60	3.50	4.90	1.30	1.20	3.10	3.6	1.2	1.2
	Equivalent Urea Quantity : 50% Consumption	Litre	2.25	2.30	0.85	0.95	1.70	1.95	0.80	0.80	1.75	2.45	0.65	0.60	1.55	1.80	0.60	0.60
	Total Eq. Fluid Consumed	Litre	74.89	76.71	41.26	37.98	78.31	81.95	41.09	43.17	73.28	75.30	38.88	36.19	76.74	80.01	38.22	40.80
	Urea uses % wrt. Diesel	%	6%	6%	4.21%	5.13%	4.44%	4.88%	3.97%	3.78%	4.89%	6.73%	3.40%	3.37%	4.12%	4.60%	3.19%	2.99%
	Fuel Economy	**kmpl**	**3.32**	**3.24**	**6.14**	**6.51**	**3.15**	**3.01**	**5.98**	**5.69**	**3.37**	**3.31**	**6.49**	**6.77**	**3.21**	**3.08**	**6.41**	**6.00**
	Status of Fuel economy	**%**	**Base**	**Base**	**Base**	**Base**	**Base**	**Base**	**Base**	**Base**	**1.55%**	**2.14%**	**5.70%**	**4.05%**	**1.89%**	**2.29%**	**7.10%**	**5.40%**
	Equivalent Fluid Economy	**kmpl**	**3.22**	**3.14**	**6.01**	**6.35**	**3.08**	**2.94**	**5.87**	**5.58**	**3.29**	**3.20**	**6.38**	**6.66**	**3.14**	**3.01**	**6.31**	**5.91**
	Status of Equivalent Fluid economy	**%**	**Base**	**Base**	**Base**	**Base**	**Base**	**Base**	**Base**	**Base**	**2.20%**	**1.87%**	**6.12%**	**4.95%**	**2.05%**	**2.42%**	**7.51%**	**5.81%**
	Cluster Total Odo. Reading (at End of trial)	km	239	239	245	237	239	239	237	237	241	241	256	244	237	237	239	239
	Cluster Trip AFE	kmpl	3.40	3.30	6.90	7.00	3.20	3.20	6.20	6.10	3.40	3.40	6.70	6.90	3.30	3.40	6.10	6.00
	Status of Cluster Trip AFE wrt Actual fuel economy	%	**2.48%**	**1.89%**	**12.43%**	**7.56%**	**1.72%**	**6.22%**	**3.65%**	**7.24%**	**0.91%**	**2.78%**	**3.28%**	**1.90%**	**2.96%**	**10.34%**	**-4.78%**	**0.08%**
Avg.kmpl Results	**Average Fuel Economy**	**kmpl**	**3.28**		**6.32**		**3.08**		**5.83**		**3.34**		**6.63**		**3.14**		**6.20**	
	Status of Average Fuel economy	**%**	**Base**		**Base**		**Base**		**Base**		**1.84%**		**4.85%**		**2.08%**		**6.27%**	
	Average Eq. Fluid Economy	**kmpl**	**3.18**		**6.18**		**3.01**		**5.72**		**3.24**		**6.52**		**3.08**		**6.11**	
	Status of Average Eq. Fluid economy	**%**	**Base**		**Base**		**Base**		**Base**		**2.04%**		**5.52%**		**2.23%**		**6.68%**	
Oveall Results L+UL	**Overall Avg. Fuel Economy (kmpl) : L + UL**	**80%** / **20%**	**3.63**				**3.40**				**3.71**				**3.49**			
	Status of Overall Avg. Fuel Economy (%)		**Base**				**Base**				**2.18%**				**2.55%**			
	Overall Avg. Eq. Fluid Economy (kmpl) : L + UL	**80%** / **20%**	**3.52**				**3.32**				**3.61**				**3.42**			
	Status of Overall Avg. Eq. Fluid Economy (%)	**%**	**Base**				**Base**				**2.42%**				**2.73%**			
	SIGNA 4221.S 5L BSVI OBD-2 Extrapolated value w/o THU bearing : (0.25% per hub (0.25*10 =2.5%) has been reduced from actual fuel/fluid economy value)																	
Avg.kmpl Results	**Average Fuel Economy with THU**	**kmpl**	**3.28**		**6.32**		**3.08**		**5.83**		**3.26**		**6.46**		**3.06**		**6.05**	
	Status of Average Fuel economy	**%**	**Base**		**Base**		**Base**		**Base**		**-0.70%**		**2.23%**		**-0.47%**		**3.61%**	
	Average Eq. Fluid Economy	**kmpl**	**3.18**		**6.18**		**3.01**		**5.72**		**3.16**		**6.36**		**3.00**		**5.95**	
	Status of Average Eq. Fluid economy	**%**	**Base**		**Base**		**Base**		**Base**		**-0.51%**		**2.88%**		**-0.32%**		**4.01%**	
Oveall Results L+UL	**Overall Avg. Fuel Economy (kmpl) : L + UL**	**80%** / **20%**	**3.63**				**3.40**				**3.61**				**3.40**			
	Status of Overall Avg. Fuel Economy (%)		**Base**				**Base**				**-0.38%**				**-0.01%**			
	Overall Avg. Eq. Fluid Economy (kmpl) : L + UL	**80%** / **20%**	**3.52**				**3.32**				**3.52**				**3.33**			
	Status of Overall Avg. Eq. Fluid Economy (%)	**%**	**Base**				**Base**				**-0.14%**				**0.16%**			

Note: Equivalent Urea consumption is taken as 50% of actual urea consumption.

8.3 Tractive Effort Comparison :



Remarks:

Signa 4221.T Cx BS-VI OBD-II (204 HP) Test vehicle tractive effort was at par in all gears as compared with Signa 4221.T BS-VI OBD-I (200 HP) _ Base vehicle.

8.4 Torque Table :

8.3.1 : Base vehicle (Signa 4221.T BS-VI OBD-I (200 HP))

FE Switch Torque Table (Torque Limit)										
	Load Gear	1st Gear	2nd Gear	3rd Gear	4th Gear	5th Gear	6th Gear	7th Gear	8th Gear	Reverse
Economy Mode	-	850	850	700	590	590	590	-	-	850
Balance Mode	-	850	850	700	700	700	700	-	-	850
Power Mode	-	850	850	850	850	850	850	-	-	850

8.3.2: Test vehicle (Signa 4221.T BS-VI OBD-II (204 HP))

FE Switch Torque Table (Torque Limit)										
	Load Gear	1st Gear	2nd Gear	3rd Gear	4th Gear	5th Gear	6th Gear	7th Gear	8th Gear	Reverse
Economy Mode	NA	850	850	700	590	590	590	-	-	850
Balance Mode	NA	850	850	700	700	700	700	-	-	850
Power Mode	NA	850	850	850	850	850	850	-	-	850

9.0 Observation: Nil

10.0 Further Testing: Further fuel economy evaluation needs to be done with benchmark vehicle to get the PALS attribute wrt core competitor.

Annexure-1

Level 1 Attribute:	PANEL Chart	VE CVBU/FE/01
PAT- Fuel Economy	Specification:Max Power- 204 hp @ 2200 rpm Max Torque - 850 Nm 1000-1600 rpm Driveline : G-1150 DD | 5.85 RAR | 295/90R 20 Tyres | 3 Mode FE Switch | Torque Factor: NA EOL Parameters : RSG: 80 kmph	Panel Chart No: VE CVBU/ HCV/ SIGNA 4221.T 5L Cx BSVI RDE/ Cx / FE/01

Programme / MY : SIGNA 4221.T Cx 5L BSVI RDE	**Gateway :** DR3	**Owner:** Ajaykumar Jadhav	**Project Name: SIGNA 4221.T 5L BSVI RDE(Cx Version)**	**Date:** 28/06/2023

Level 1 Summary:

	PALS Position	Status: Current/ Gateway	Status: Job#1 Predicted
Level 1:	L	L	

Statement Of Intent (SOI) or DNA Description:

Fuel Economy: Haulage Application

Attribute Target Summary:

Level 2 / Level 3 / Level 4 / Level 5 / Level 6	PALS	Metric	Target	Status: Current/Gateway SIGNA 4221.T Cx BSVI RDE — With THU bearing	Without THU bearing Extrapolated	Job#1 Predicted	Base Vehicle: Base Signa 4221.T BSVI	Remarks
Specification							--	
Vehicle GVW (Payload)		kg		42000			42000	
Power (Max) @ rpm		HP		204 hp @ 2200 rpm			204 hp @ 2200 rpm	
Power to Weight ratio		HP/Ton		4.86			4.86	
Torque (Max) @ rpm		Nm		850 Nm @ 1000-1600 rpm,			850 Nm @ 1000-1600 rpm	
Torque to Weight ratio		Nm/Ton		20.24			20.24	
Tyres size		--		JK Tyre LCCR 295/90R20			JK Tyre non LCCR 295/90R20	
Gear Box Ratios		--		G950 DD, FGR -9.13			G950 DD, FGR -9.13	
Final Drive Ratio		--		5.85			5.85	
Fluid Economy at Rated Load	L			L	L			
Highway AC OFF		--						
Route : Jamshedpur-Ranchi		--						
Avg. Fluid Economy @40 kmph		kmpl	**≥ 3.18**	**3.24**	**3.18**		3.18	*Signa 4221.T OBII Cx and Signa 4221.T OBDI was tested back to back condition 0n JSR-Ranchi route.
Avg. Fluid Economy @55 kmph		kmpl	**≥ 3.01**	**3.08**	**3.01**		3.01	
Fluid Economy At Unladen	L			L	L			
Highway AC OFF		--						
Route : Jamshedpur-Ranchi		--						
Avg. Fluid Economy @40 kmph		kmpl	**≥ 6.18**	**6.52**	**6.36**		6.18	*Signa 4221.T OBII Cx and Signa 4221.T OBDI was tested back to back condition 0n JSR-Ranchi route.
Avg. Fluid Economy @55 kmph		kmpl	**≥ 5.72**	**6.11**	**5.96**		5.72	
Overall Fuel Economy (80% Laden + 20% Unladen)	L			L	L			
Highway AC OFF		--						
Route : Jamshedpur-Ranchi		--						
Overall Avg. Fluid Economy @40 kmph		kmpl	**≥ 3.52**	**3.61**	**3.52**		3.52	*Signa 4221.T OBII Cx and Signa 4221.T OBDI was tested back to back condition 0n JSR-Ranchi route.
Overall Avg. Fluid Economy @55 kmph		kmpl	**≥ 3.32**	**3.42**	**3.33**		3.32	

Executive Summary:

1. Test vehicle SIGNA 4221.T Cx BSVI OBD-2 tested with : 204 HP Engine calibration with 95% Cal Maturity + THU Bearing + Low viscous 5W30 engine oil + Low viscous 80W90 pro FE rear axle oil +JK LCRR tyres + 24" Fan
2. Target has been taken at-par (As per Cx approach) wrt to base SIGNA 4221.T OBD-1 BSVI vehicle.
3. SIGNA 4221.T Cx RDE BSVI is meeting the PALS target of "Leadership".

4.Concluded fluid economy results is extrapolated for Cx variant to reducing the THU bearing contribution by 2.5% from actual fluid economy results.

Tested by:	Verified and Approved by:
Palash Das/Debabrata Das / Swapnil Jadhav **PAT ENGINEER**	Ajaykumar Jadhav / Sanjeev Kumar **PAT LEAD / PROJECT LEAD**

Enclosed with FIR/Testing-Outdoor/224000/2023/6205

Annexure-2

TATA

TATA MOTORS LTD.
ERC,PUNE

PAT ATTRIBUTES SHEET - Fuel Economy

VE CVBU/FE/01

PAT Sheet No.: VE CVBU/ HCV/ SIGNA 4221.T 5L BSVI RDE/ Cx / FE/01

Depart ment Name: Vehicle Engineering

Project Name: SIGNA 4221.T 5L BSVI RDE_Cx (G-1150 DD FGR-9.13, 5.85 RAR, 295/90R 20 phase II low CRR Tyres, 3 Mode FE Switch)
Base Vehicle: SIGNA 4221.T 5L BSVI OBD-1_G-1150 DD, FGR-9.13, 5.85 RAR, 295/90R 20 Tyres_3 Mode FE Switch
Application: Haulage

Date: 28/06/2023

Sr.No.	Aggregate and test parameter	Unit	Test Type	Classification @DR4	Ref.Standard / Work Guideline	Simulation	Base Vehicle SIGNA 4221.T BSVI OBD-1 (G 1150 DD | RAR 5.85 | 295/90R20)	Target	Test Vehicle SIGNA 4221.T BSVI OBD-II Cx (G 1150 DD | RAR 5.5 | 295/90R20)		Status (RYG)	Remarks	CoC Remark
	Level 2 / Level 3 / Level 4 / Level 5 / Level 6 / Level 7												
A	**Fuel Economy Evaluation**			Heading			--		--		--		
2.0	Fuel Economy At Rated Load			Heading									
2.7	Highway AC OFF			Heading					with THU Bearing	without THU Bearing			
2.7.1	**Route : Jamshedpur-Ranchi**			Specification								1. Test vehicle SIGNA 4221.T Cx BSVI OBD-2 tested with : 204 HP Engine calibration with 95% Cal Maturity + THU Bearing + Low viscous 5W30 engine oil + Low viscous 80W90 pro FE rear axle oil + Bridgestone LCRR tyres + 24" Fan 2. Target has been taken at-par (As per Cx approach) wrt to base SIGNA 4221.T OBD-1 BSVI vehicle. 3. SIGNA 4221.T Cx RDE BSVI is not meeting the PALS target of "Leadership". 4.Concluded fluid economy results is extrapolated for Cx variant to reducing the THU bearing contribution by 2.5% from actual fluid economy results.	
2.7.9	Average Fuel Economy @40 kmph	Kmpl		Data Genration			3.28	--	3.34	3.26	--		
2.7.10	Average Fuel Economy @55 kmph	Kmpl	Standard	Data Genration	TST/VT/WG/158		3.08	--	3.14	3.06	--		
2.7.11	**Average Fluid Economy @40 kmph**	Kmpl		Target			**3.18**	≥ **3.18**	**3.24**	**3.18**	G		
2.7.12	**Average Fluid Economy @55 kmph**	Kmpl		Target			**3.01**	≥ **3.01**	**3.08**	**3.01**	G		
4.0	Fuel Economy At Unladen			Heading									
4.7	Highway AC OFF			Heading									
4.7.1	**Route : Jamshedpur-Ranchi**			Specification								1. Test vehicle SIGNA 4221.T Cx BSVI OBD-2 tested with : 204 HP Engine calibration with 95% Cal Maturity + THU Bearing + Low viscous 5W30 engine oil + Low viscous 80W90 pro FE rear axle oil + Bridgestone LCRR tyres + 24" Fan 2. Target has been taken at-par (As per Cx approach) wrt to base SIGNA 4221.T OBD-1 BSVI vehicle. 3. SIGNA 4221.T Cx RDE BSVI is not meeting the PALS target of "Leadership". 4.Concluded fluid economy results is extrapolated for Cx variant to reducing the THU bearing contribution by 2.5% from actual fluid economy results.	
4.7.9	Average Fuel Economy @40 kmph	Kmpl		Data Genration			6.32	--	6.63	6.46	--		
4.7.10	Average Fuel Economy @55 kmph	Kmpl	Standard	Data Genration	TST/VT/WG/158		5.83	--	6.20	6.05	--		
4.7.11	**Average Fluid Economy @40 kmph**	Kmpl		Target			**6.18**	≥ **6.18**	**6.52**	**6.36**	G		
4.7.12	**Average Fluid Economy @55 kmph**	Kmpl		Target			**5.72**	≥ **5.72**	**6.11**	**5.96**	G		
5.0	Overall Fuel Economy (80% Laden + 20% Unladen)			Heading									
5.1	Highway AC OFF			Heading									
5.1.1	**Route : Jamshedpur-Ranchi**			Specification								1. Test vehicle SIGNA 4221.T Cx BSVI OBD-2 tested with : 204 HP Engine calibration with 95% Cal Maturity + THU Bearing + Low viscous 5W30 engine oil + Low viscous 80W90 pro FE rear axle oil + Bridgestone LCRR tyres + 24" Fan 2. Target has been taken at-par (As per Cx approach) wrt to base SIGNA 4221.T OBD-1 BSVI vehicle. 3. SIGNA 4221.T Cx RDE BSVI is not meeting the PALS target of "Leadership". 4.Concluded fluid economy results is extrapolated for Cx variant to reducing the THU bearing contribution by 2.5% from actual fluid economy results.	
5.1.9	Average Overall Fuel Economy @40 kmph	Kmpl		Data Genration			3.63	--	3.71	3.62	--		
5.1.10	Average Overall Fuel Economy @55 kmph	Kmpl		Data Genration	TST/VT/WG/158		3.40	--	3.49	3.40	--		
5.1.11	**Overall Average Fluid Economy @40 kmph**	Kmpl		Target			**3.52**	≥ **3.52**	**3.61**	**3.52**	G		
5.1.12	**Overall Average Fluid Economy @55 kmph**	Kmpl		Target			**3.32**	≥ **3.32**	**3.42**	**3.33**	G		

Sign Off

Responsibility	Name	Signature
PAT Engineer	Palash Das /Debabrata Das/ Swapnil Jadhav	
PAT Lead / Project Lead	Ajaykumar Jadhav / Sanjeev kumar	

Index for Status

Off target, no resolution
Marginally off target, resolution planned/low risk
Met target, no concern

Enclosed with FIR/Testing-Outdoor/224000/2023/6205

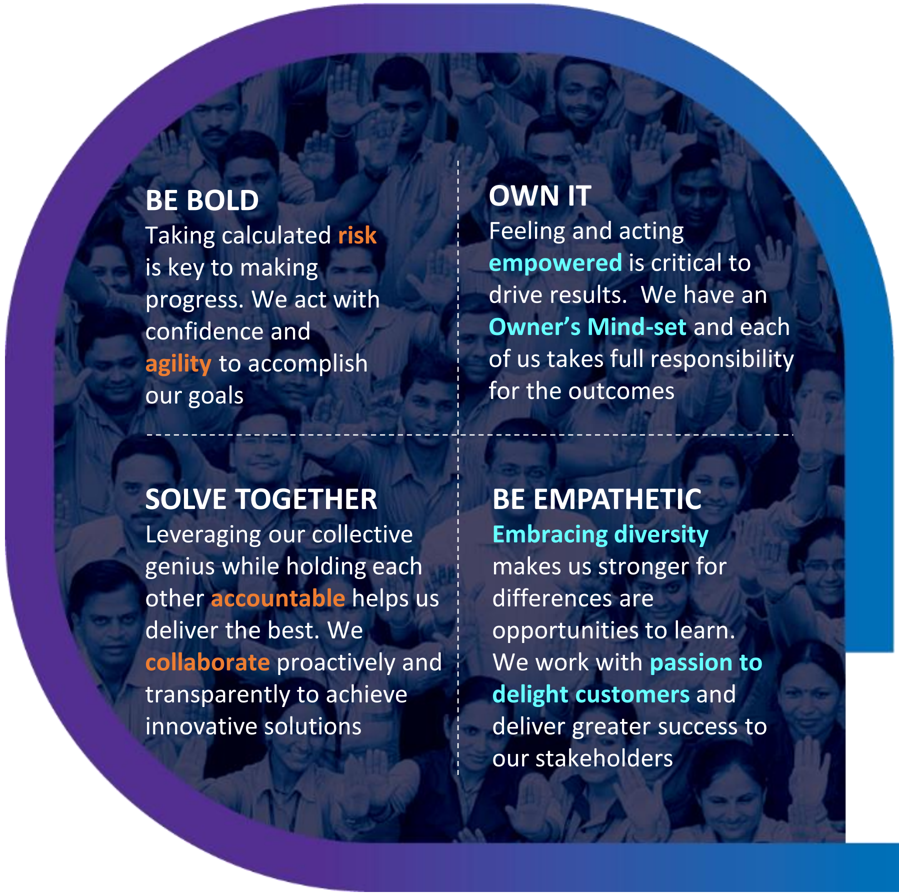
TATA MOTORS
Connecting Aspirations

Fuel Economy Evaluation on Signa 4221.T 5L RDE BSVI with Signa 4221.T 5L BSVI Base Vehicle.

PED - HCV I 28.06.2023

OUR CULTURE CREDO

AT TATA MOTORS
We are connecting aspirations by being bold in thought and action, owning every opportunity and challenge, Solving together as one team and engaging all our stakeholders with empathy.
We are **MORE WHEN ONE!**



MORE WHEN ONE
BE BOLD | OWN IT | SOLVE TOGETHER | BE EMPATHETIC

Annexure 3

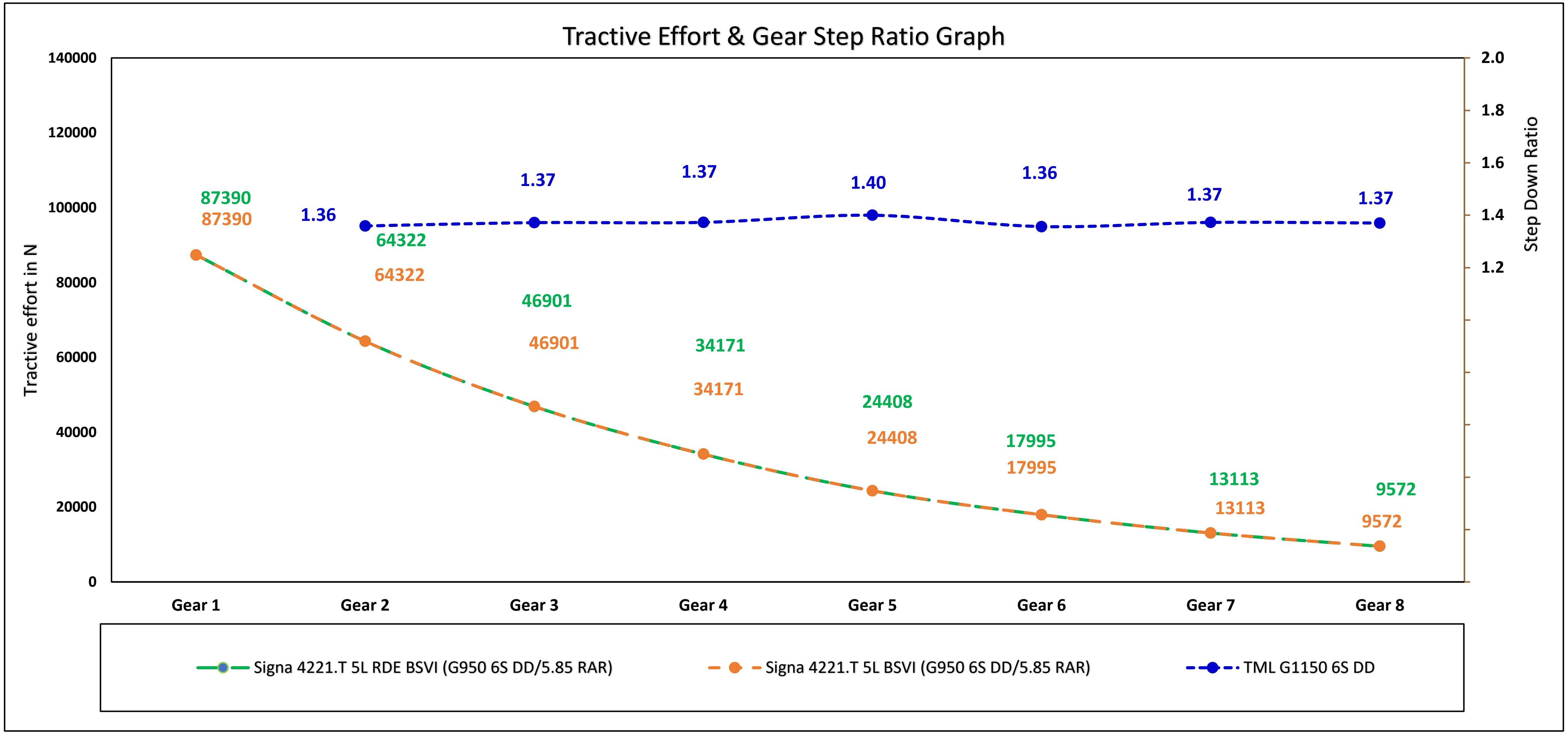
Vehicle Specification

TATA MOTORS
Connecting Aspirations

Specification		Test Vehicle	Base Vehicle
Vehicle	Model	Signa 4221.T TML5L BSVI RDE	Signa 4221.T TML5L BSVI
	GVW (Rated load) Kg	42000(Kg)	42000(kg)
	GVW (Unladen) kg	--	--
Engine	Emission norm	BSVI RDE	BSVI
	Engine	TML 5LNGE	TML 5LNGE
	Calibration	95%	As per production released
	Technology	--	--
	Torque	850 Nm @ 1000-1600 rpm	850 Nm @ 1000 - 1600 rpm
	Power	204 HP @ 2200 rpm	200 HP @ 2200 rpm
	Torque to weight ratio	20.24	20.24
	Power to weight ratio	4.86	4.76
	Torque band	600	600
Driveline	Gear box	G1150 6S DD, FGR- 9.13	G1150 6S DD, FGR- 9.13
	RAR	5.86	5.86
	Tyre	295/90R20	295/90R20

Tractive Effort Comparison

TATA MOTORS
Connecting Aspirations

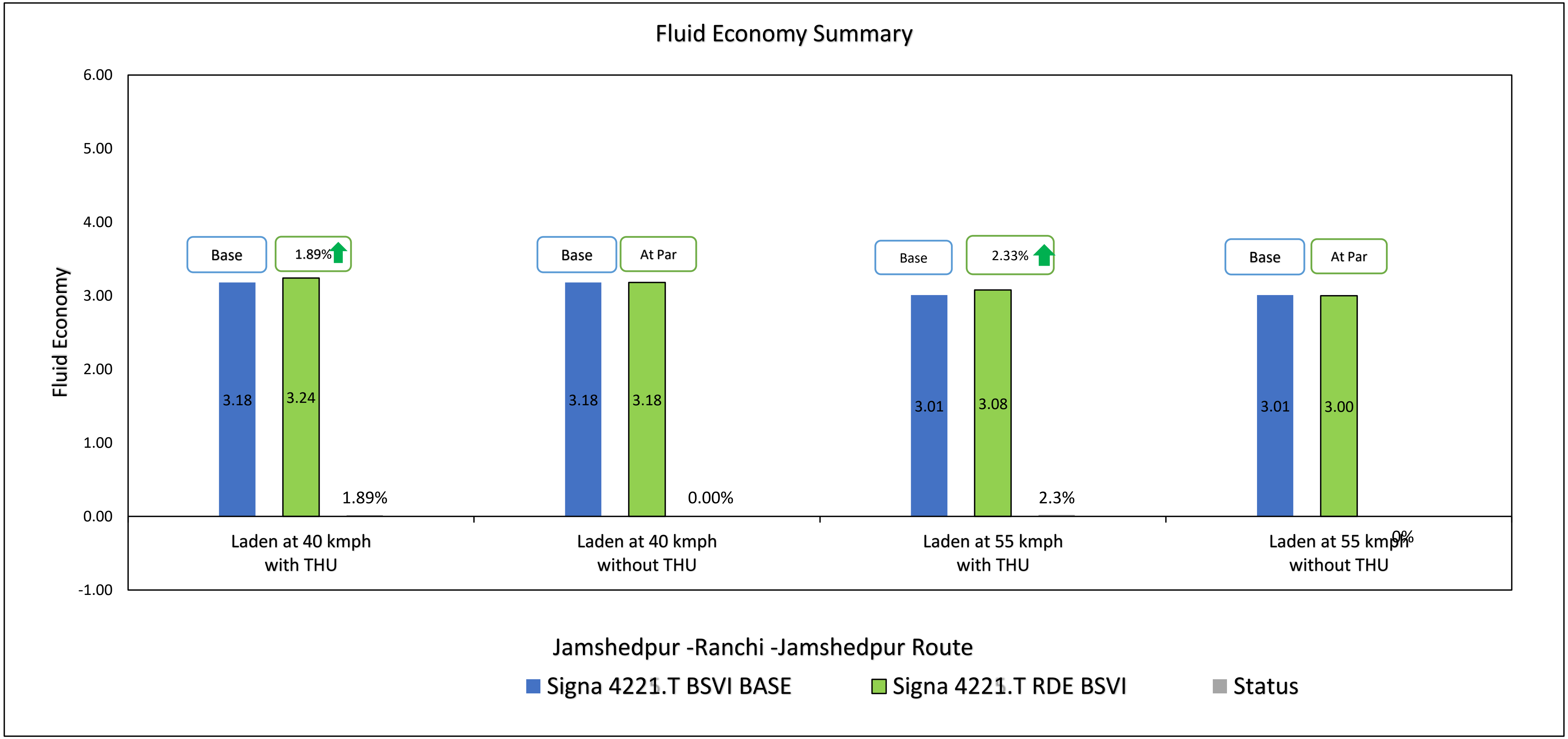


- Tractive effort of Signa 2821.T RDE vehicle is better than Signa 2821.T BSVI vehicle.

Annexure 3

Summary FE Status with all FE Avenues(Laden)

TATA MOTORS
Connecting Aspirations



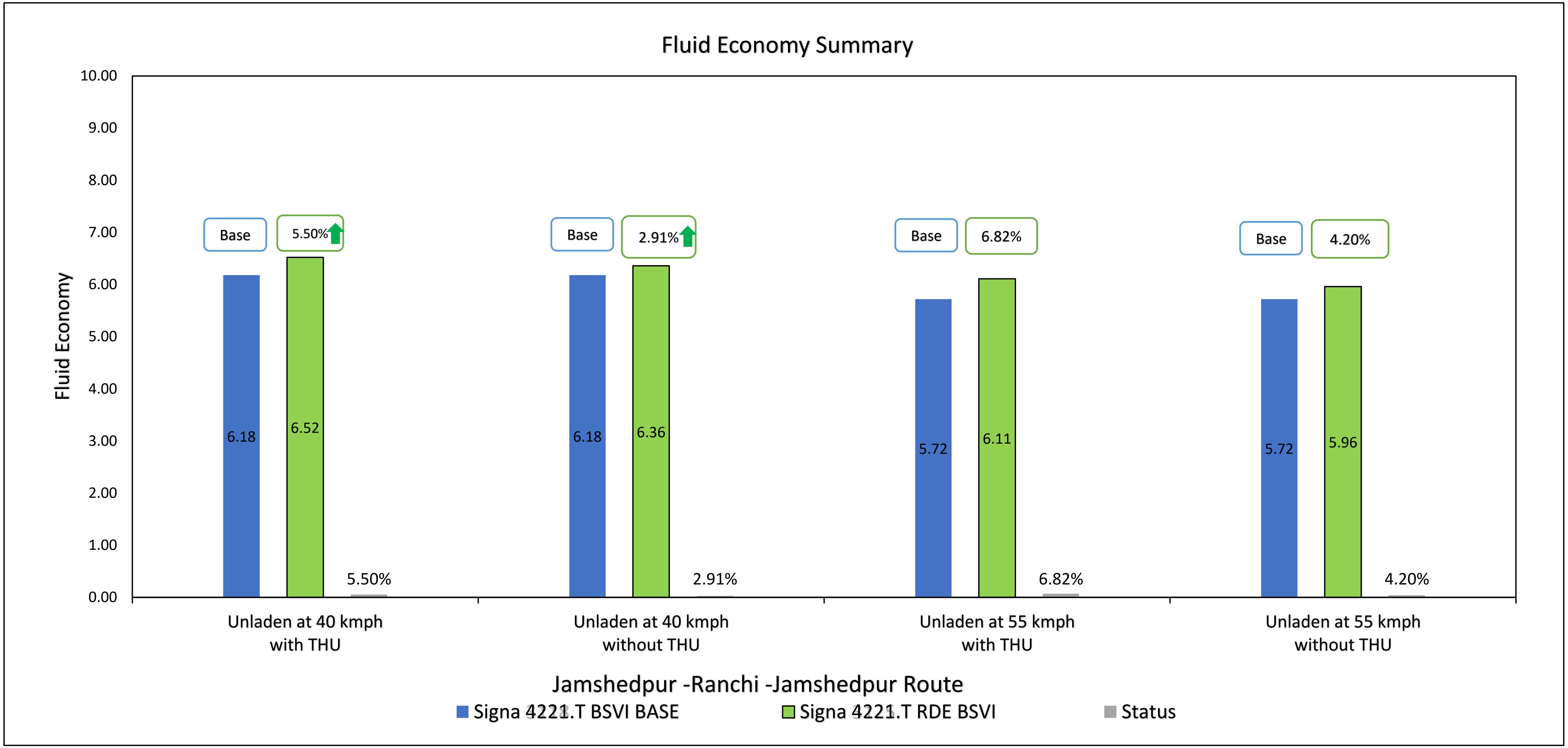
MORE WHEN ONE
BE BOLD | OWN IT | SOLVE TOGETHER | BE EMPATHETIC

Enclosed with FIR/Testing-Outdoor/224000/2023/6205

Annexure 3

Summary FE Status with all FE Avenues(Unladen)

TATA MOTORS
Connecting Aspirations



Duty Cycle Analysis

Jamshedpur-Ranchi Road FE Trial

Annexure 3

Duty Cycle : Summary (Laden)

TATA MOTORS
Connecting Aspirations

JSR-RANCHI-JSR Duty Cycle

Parameters		08.06.2023(40 kmph) Signa 4221.T RDE Test Vehicle	06.06.2023(55 kmph) Signa 4221.T RDE Test Vehicle
Engine idle speed (rpm)		750	750
Low idle time (%)		--	--
Average engine speed (rpm)		1227	--
Average power (kW)		46.89	--
Avg. vehicle speed (kmph)	Including Stop		44.18
	Excluding Stop	--	--
FE switch utilization (%time)	Power	37.74	42.01
	Balance	63.9	17.1
	Eco	25.3	8.5
Gear utilization (%time)	Neutral Gear	5.8	74
	6^{th} Gear	4.9	4.0
	7^{th} Gear	8.5	10.4
	8^{th} Gear	72.9	67.4
	Total shifts	--	--
Throttle (%time)	100%	98	400
	0%	39.6	49.0
Coolant Temp (°C)	80°C	32	33.8
	85°C	7.4	3.4
	90°C	29.3	15.5
SCRTM Fuel spent (%)		52.9	78.8
Regeneration Fuel spent (%)		--	--
Duty Cycle fuel economy (kmpl)		3.34	3.92
Duty cycle Distance(km)		242	242
Duty Cycle Time(Hrs)		'--	5.25
Actual Avg. Fuel Economy		3.24	3.08

Duty Cycle : Summary (Unladen)

TATA MOTORS
Connecting Aspirations

JSR-RANCHI-JSR Duty Cycle

		14.05.2023(40 kmph)	17.05.2023(55 kmph)
Parameters		Signa 4221.T RDE Test Vehicle	Signa 4221.T RDE Test Vehicle
Engine idle speed (rpm)		**750**	**750**
Low idle time (%)		--	--
Average engine speed (rpm)		1160	1503
Average power (kW)		--	--
Avg. vehicle speed (kmph)	Including Stop	34.96	45.05
	Excluding Stop	--	--
FE switch utilization (%time)	Power	0.00	0.01
	Balance	1.7	8.8
	Eco	98.3	91.1
Gear utilization (%time)	Neutral Gear	2.9	5.5
	6th Gear	1.1	1.5
	7th Gear	2.1	2.4
	8th Gear	89.9	85.7
	Total shifts	127	--
Throttle (%time)	100%	0.6	0
	0%	26.7	21.6
Coolant Temp (°C)	80°C	**67.1**	**3.4**
	85°C	**25.8**	**26.5**
	90°C	**4.6**	**67.5**
SCRTM Fuel spent (%)		--	--
Regeneration Fuel spent (%)		0.00	0.00
Duty cycle fuel economy (kmpl)		**5.63**	**5.69**
Duty cycle Distance(km)		**192**	**225**
Duty Cycle Time(Hrs)		**--**	**5.25**
Actual Avg. Fuel Economy		**6.52**	**6.11**

MOREWHEN ONE
BE BOLD | OWN IT | SOLVE TOGETHER | BE EMPATHETIC

Enclosed with FIR/Testing-Outdoor/224000/2023/6205

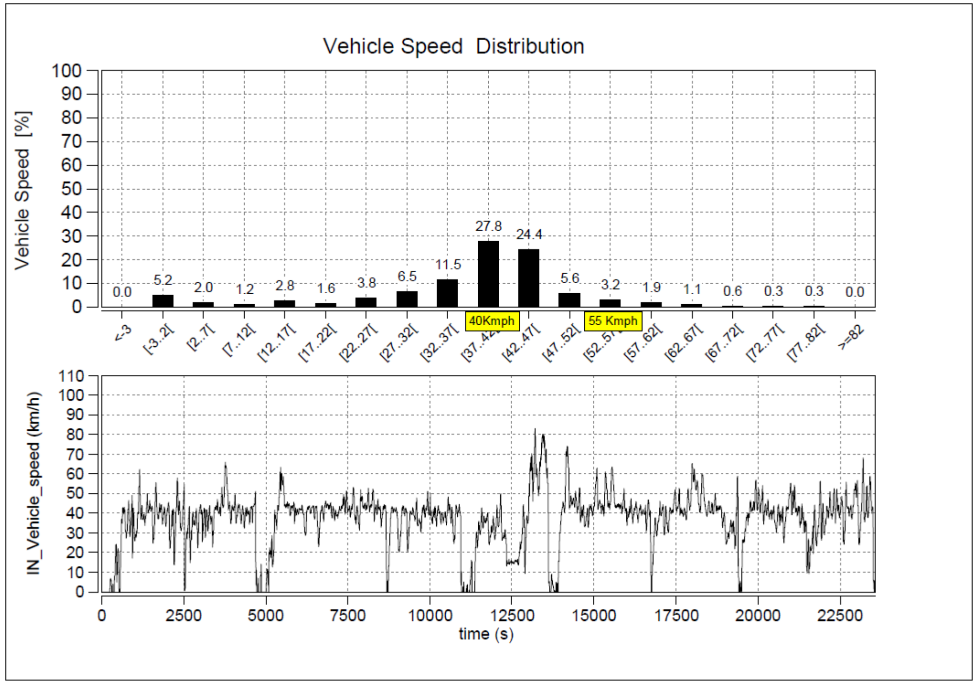
Vehicle Speed Distribution

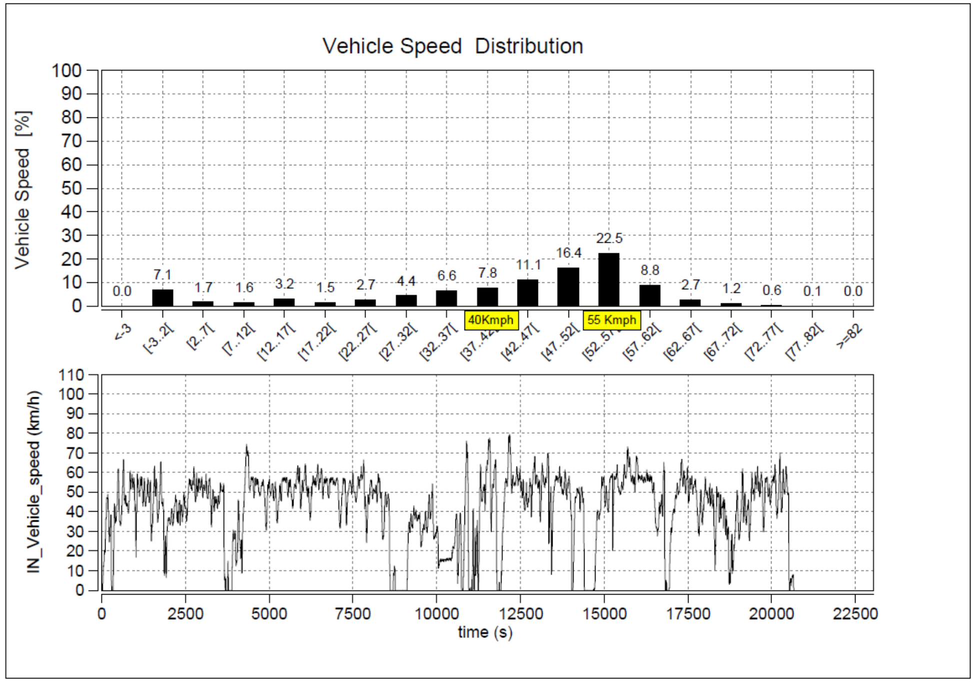
Laden Trial

TATA MOTORS
Connecting Aspirations

Signa 4221.T RDE Test Vehicle

08.06.23	40 kmph	06.06.23	55 kmph





MOREWHEN ONE
BE BOLD | OWN IT | SOLVE TOGETHER | BE EMPATHETIC

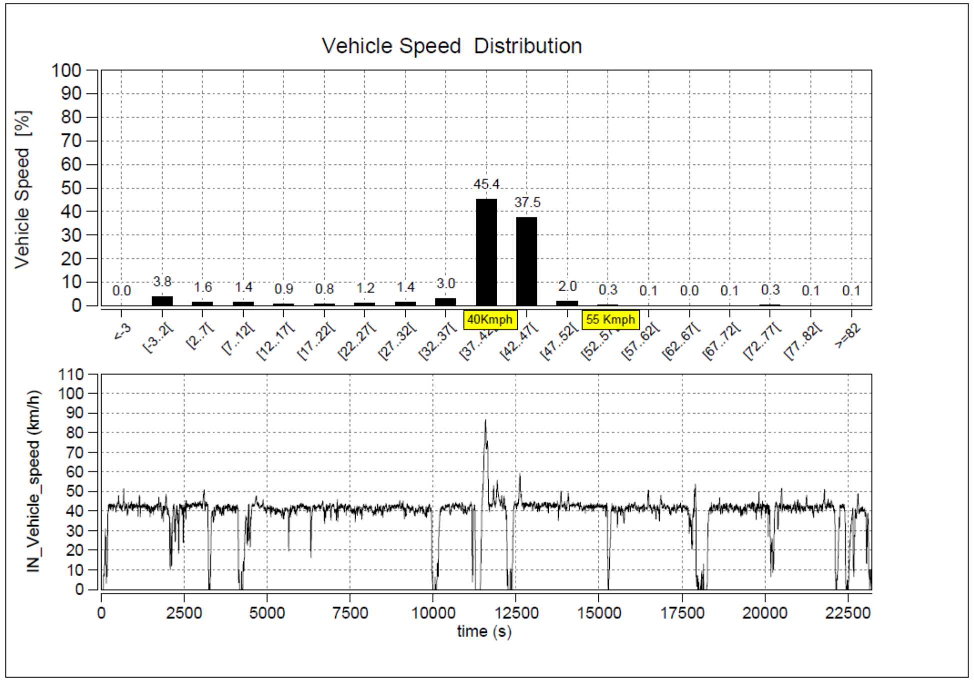
Vehicle Speed Distribution

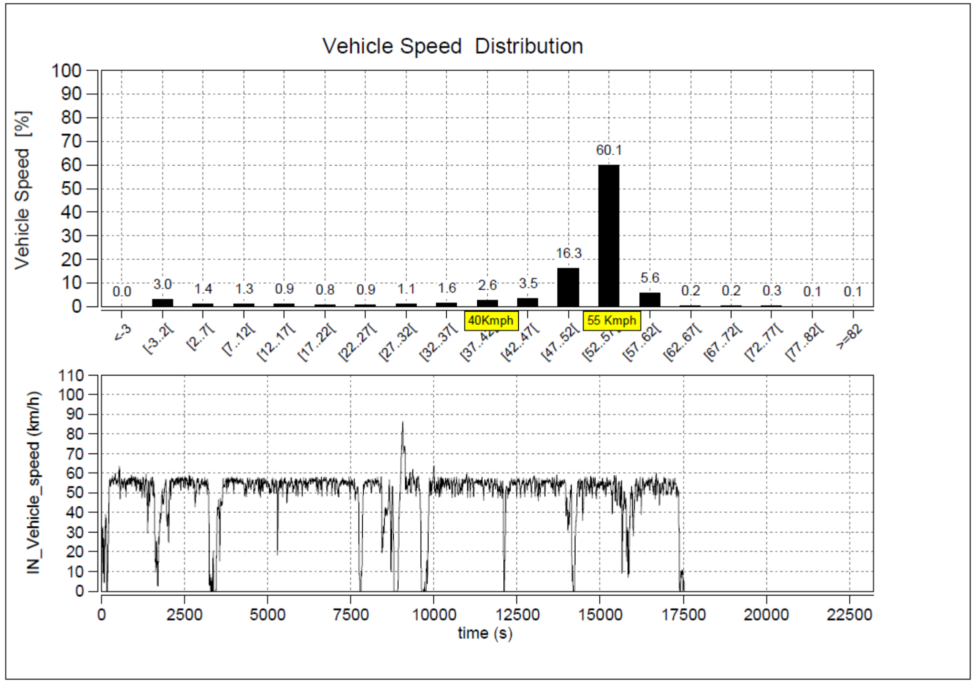
Annexure 3

Unladen Trial

Signa 4221.T RDE Test Vehicle

14.06.23	40 kmph	17.06.23	55 kmph





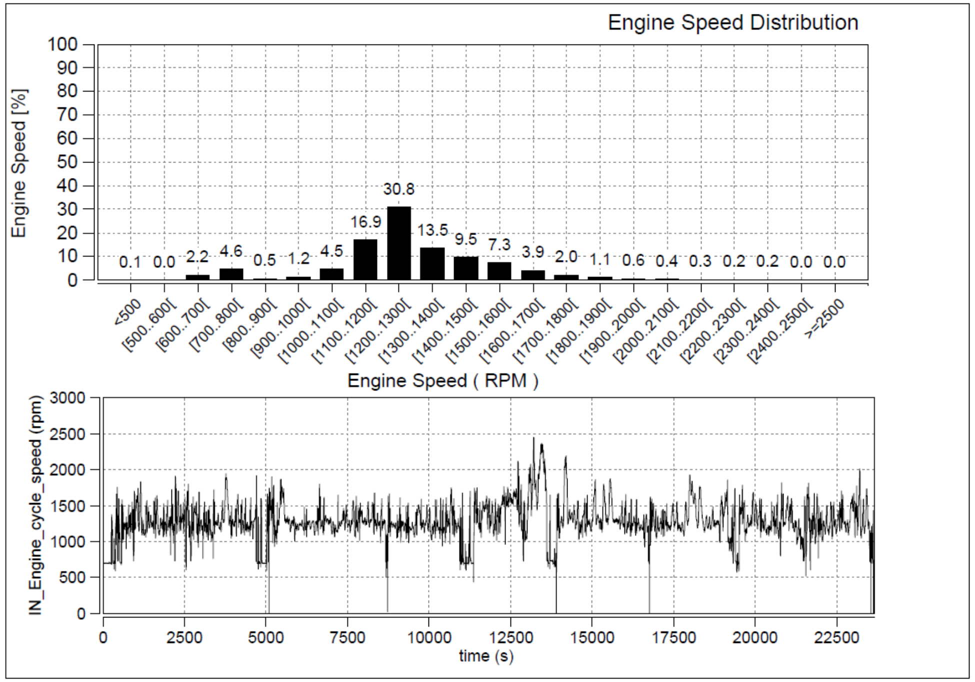
Engine Speed Distribution

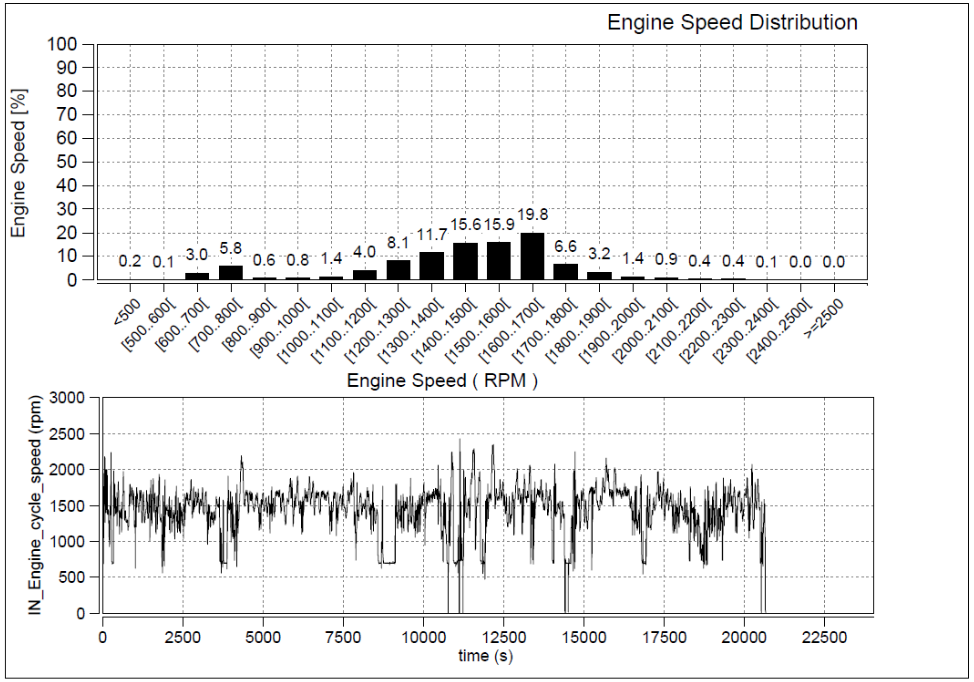
Laden Trial

TATA MOTORS
Connecting Aspirations

Signa 4221.T RDE Test Vehicle

08.06.23	40 kmph	06.06.23	55 kmph



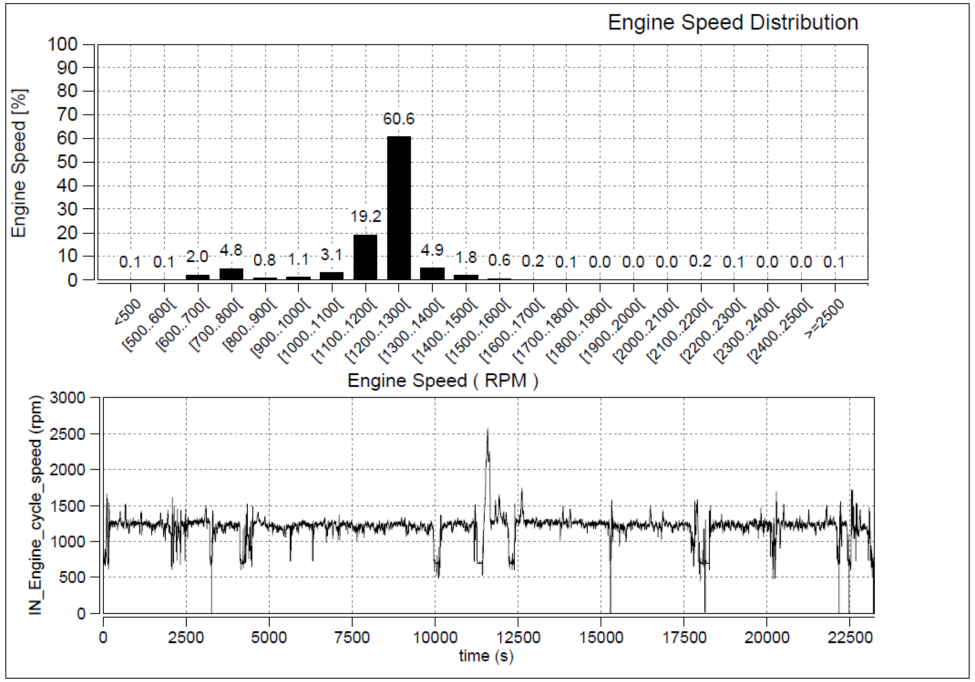


Engine Speed Distribution

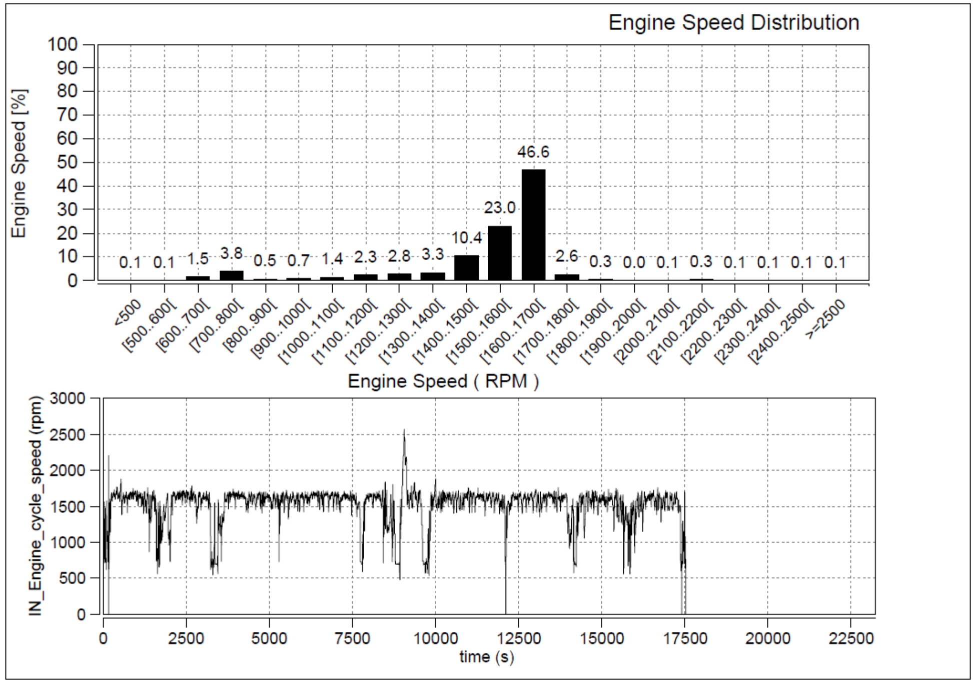
Unladen Trial

Signa 4221.T RDE Test Vehicle

14.06.23	40 kmph



17.06.23	55 kmph



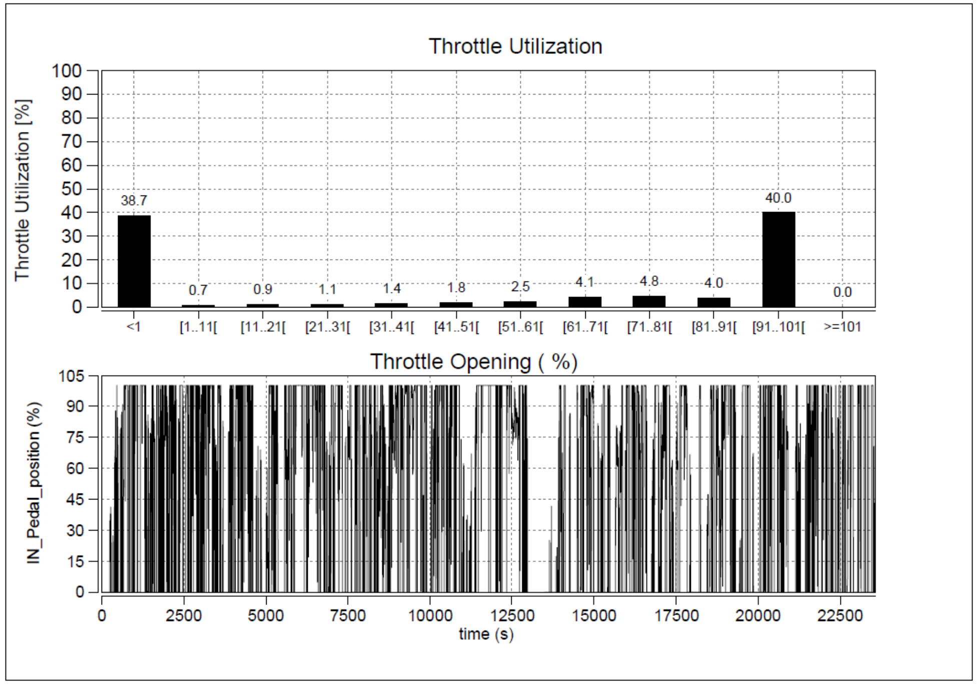
Throttle Distribution

Annexure-3

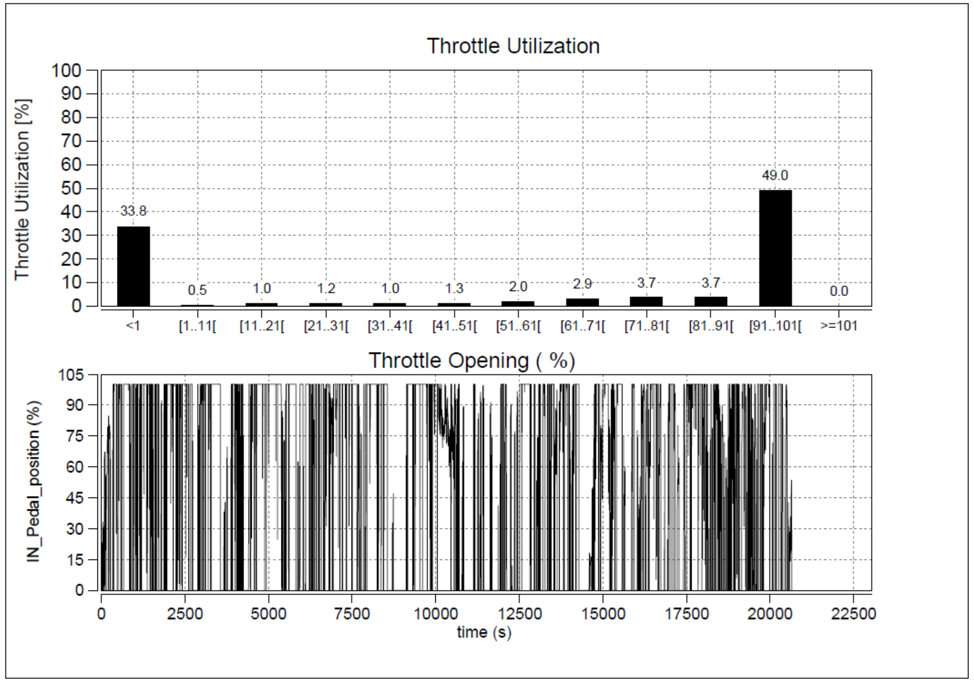
Laden Trial

Signa 4221.T RDE Test Vehicle

08.06.23	40 kmph



06.06.23	55 kmph



Enclosed with FIR/Testing-Outdoor/224000/2023/6205

Throttle Distribution

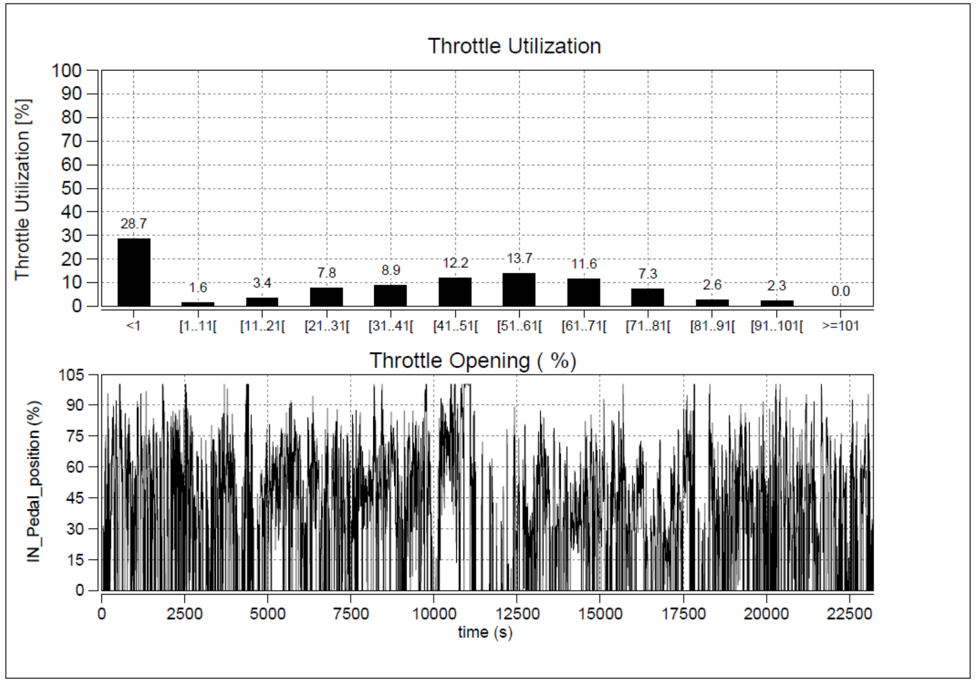
Annexure 3

TATA MOTORS
Connecting Aspirations

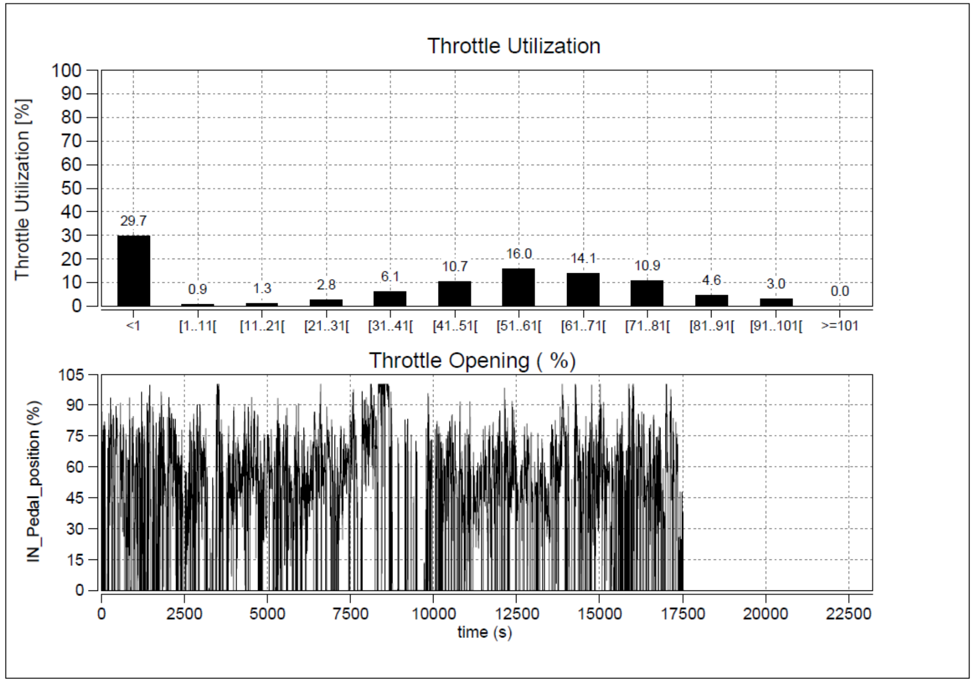
Unladen Trial

Signa 4221.T RDE Test Vehicle

14.06.23	40 kmph



17.06.23	55 kmph



MORE WHEN ONE
BE BOLD | OWN IT | SOLVE TOGETHER | BE EMPATHETIC

Enclosed with FIR/Testing-Outdoor/224000/2023/6205

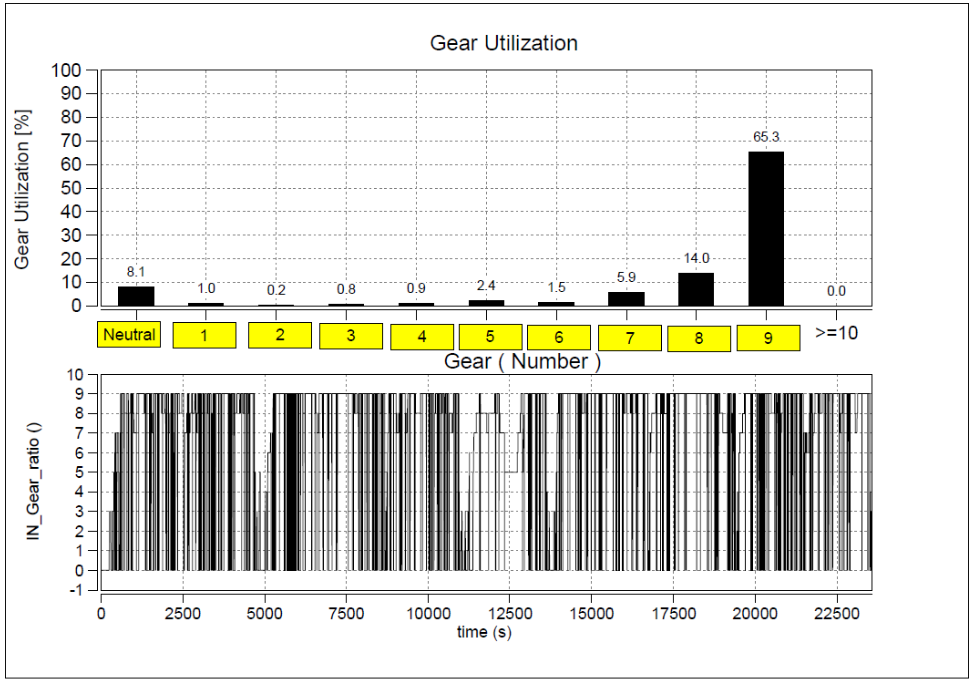
Gear Utilization

Laden Trial

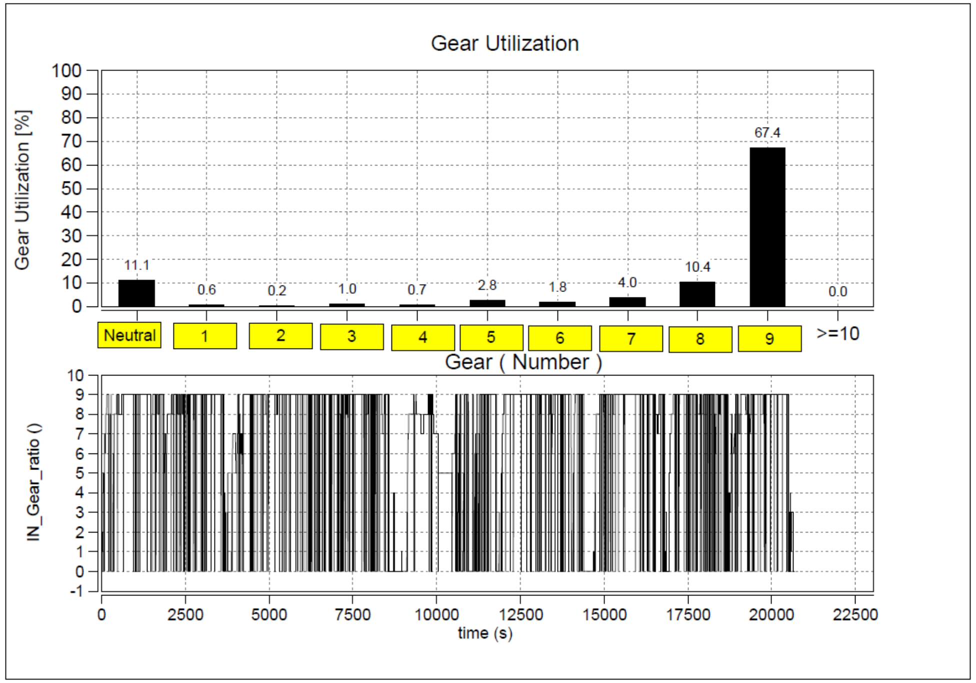
TATA MOTORS
Connecting Aspirations

Signa 4221.T RDE Test Vehicle

08.06.23 **40 kmph**



06.06.23 **55 kmph**



MOREWHEN ONE
BE BOLD | OWN IT | SOLVE TOGETHER | BE EMPATHETIC

Enclosed with FIR/Testing-Outdoor/224000/2023/6205

Annexure 3

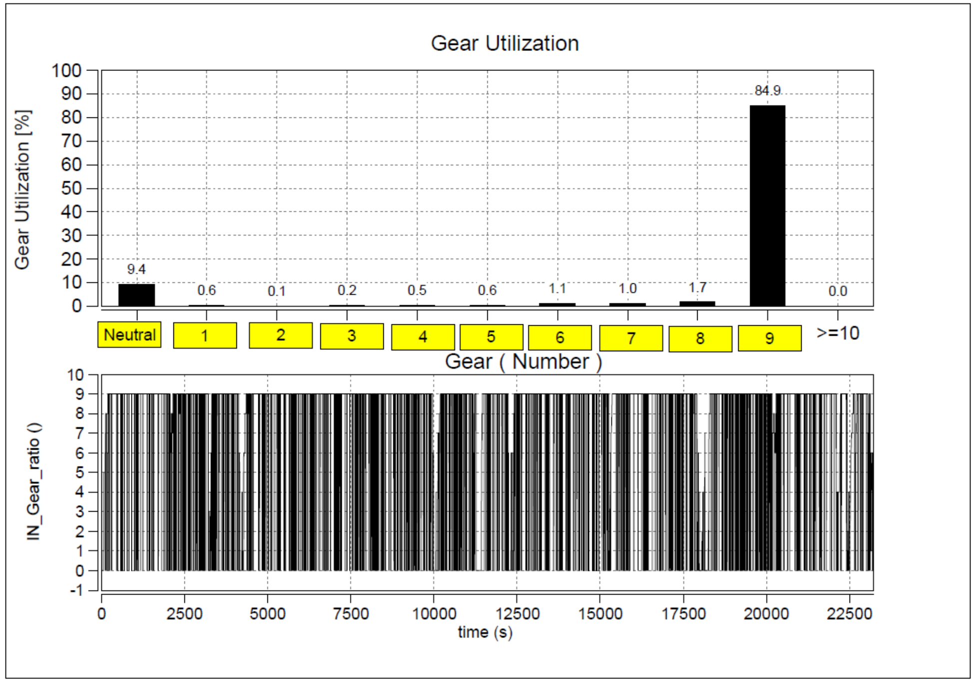
Gear Utilization

Unladen Trial

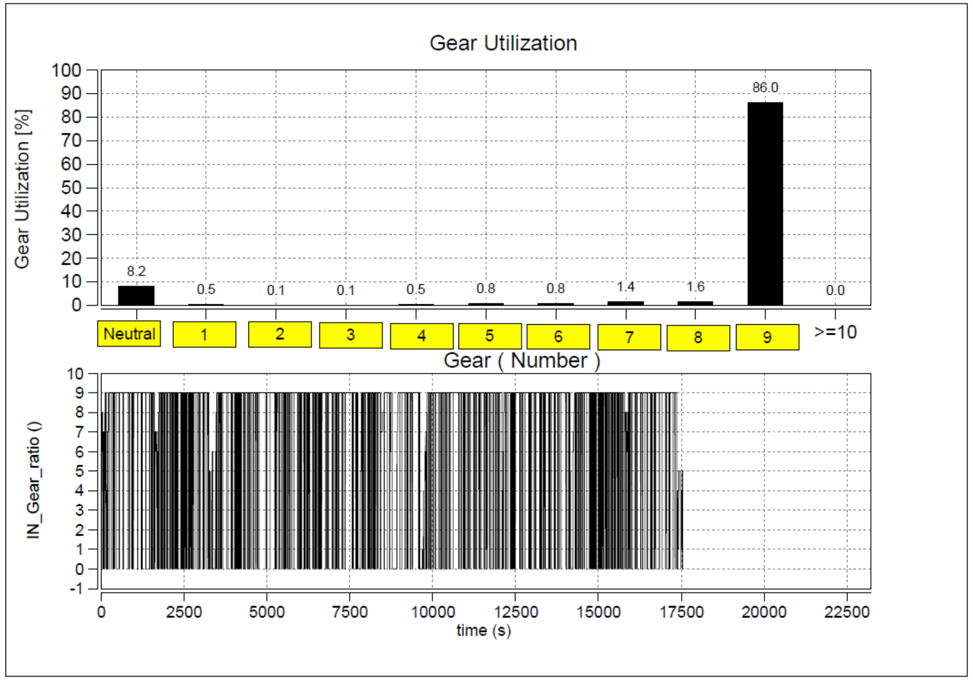
TATA MOTORS
Connecting Aspirations

Signa 4221.T RDE Test Vehicle

14.06.23 **40 kmph**



17.06.23 **55 kmph**



MORE WHEN ONE
BE BOLD | OWN IT | SOLVE TOGETHER | BE EMPATHETIC

Enclosed with FIR/Testing-Outdoor/224000/2023/6205

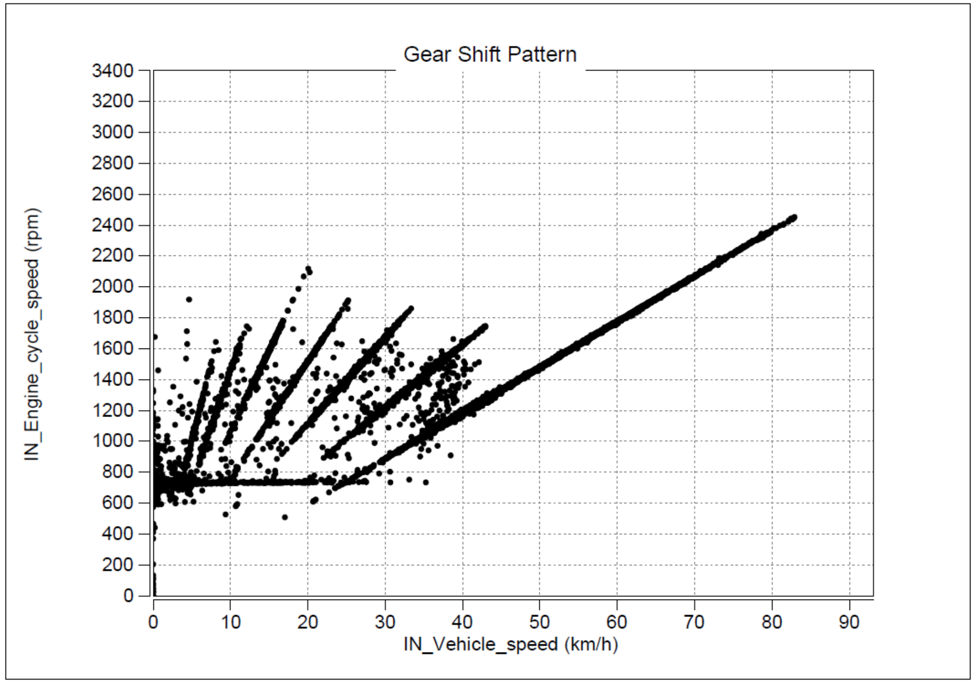
Transmission Shift Pattern

Laden Trial

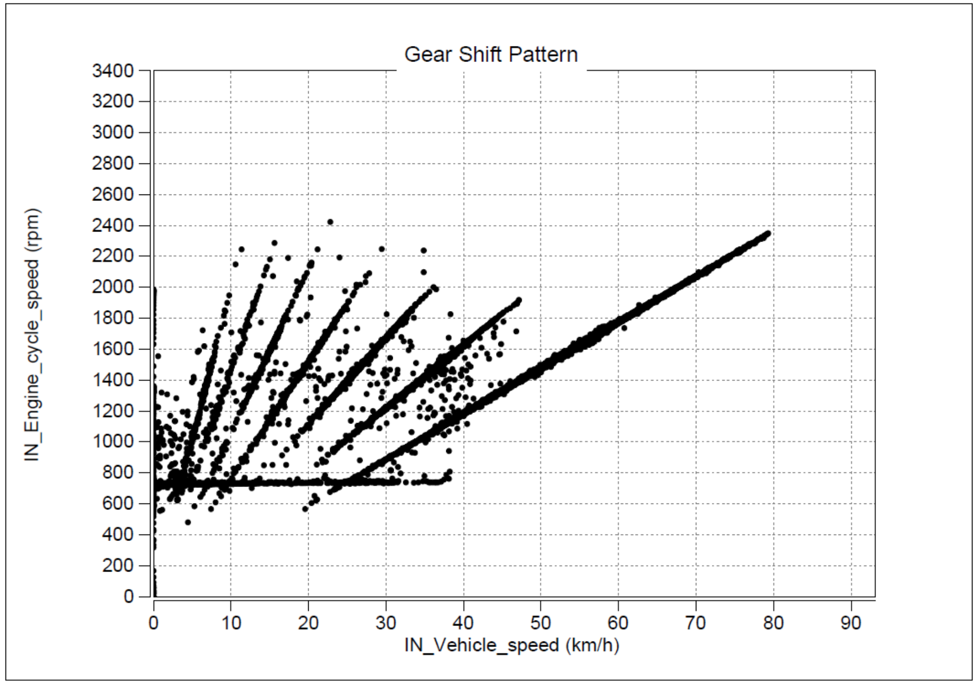
TATA MOTORS
Connecting Aspirations

Signa 4221.T RDE Test Vehicle

08.06.23	40 kmph



06.06.23	55 kmph

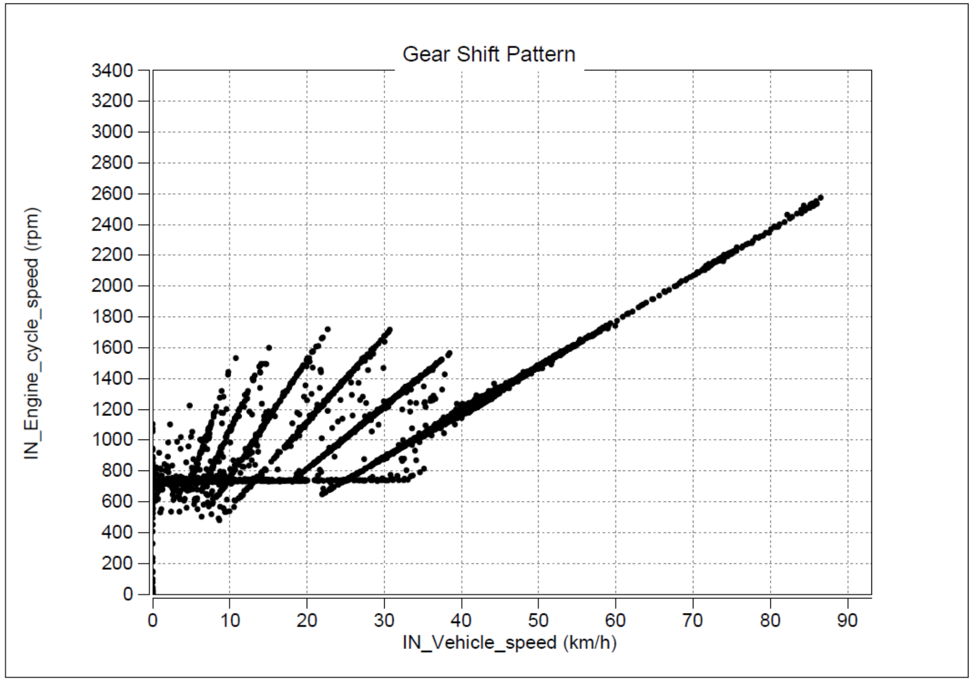


Transmission Shift Pattern

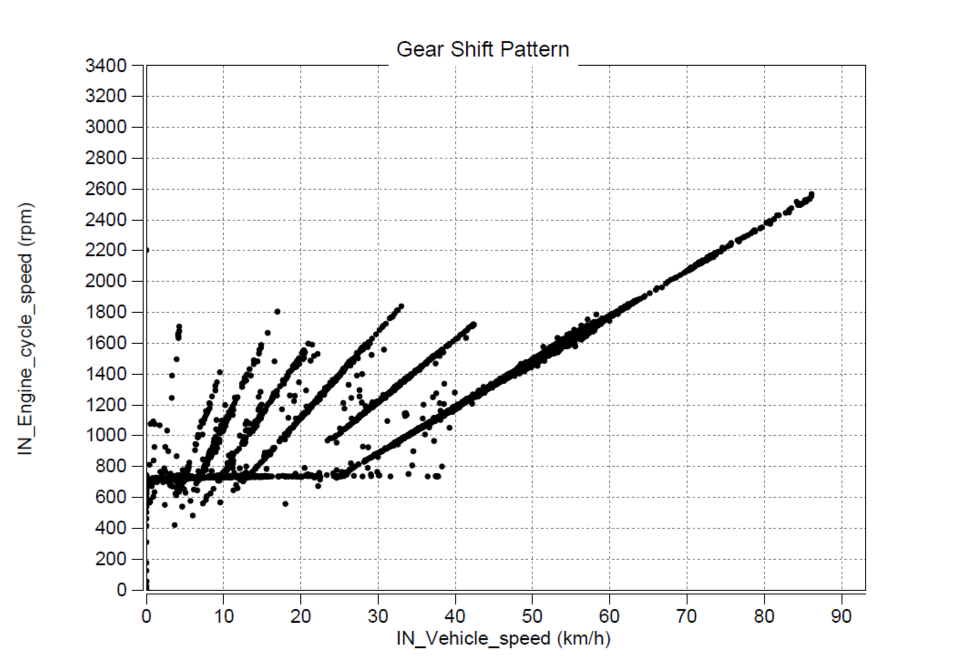
Unladen Trial

Signa 4221.T RDE Test Vehicle

14.06.23 **40 kmph**



17.06.23 **55 kmph**



Enclosed with FIR/Testing-Outdoor/224000/2023/6205

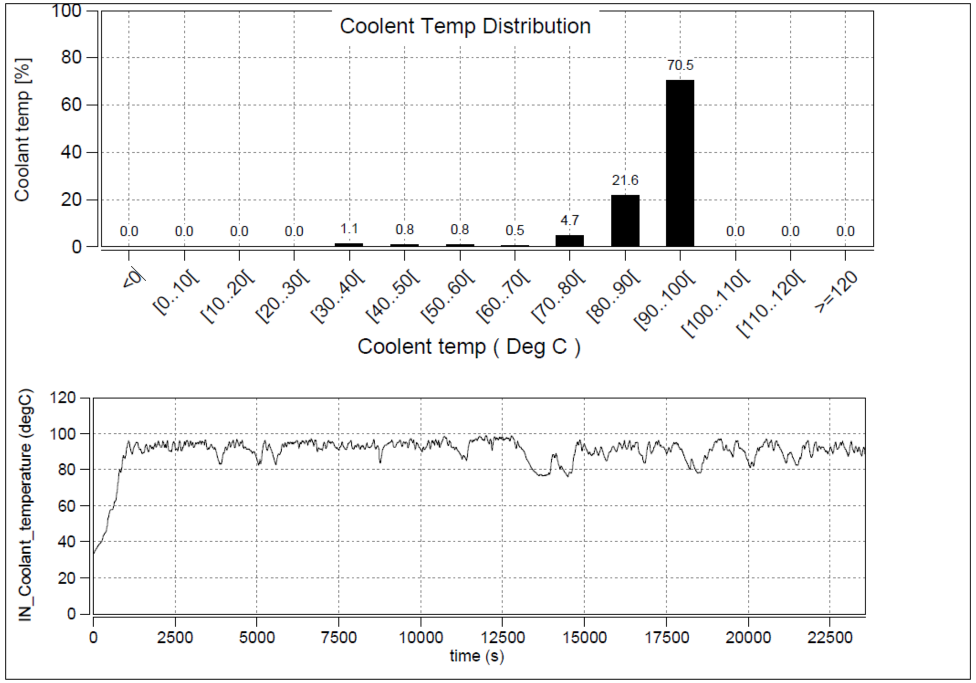
Coolant Temp Distribution

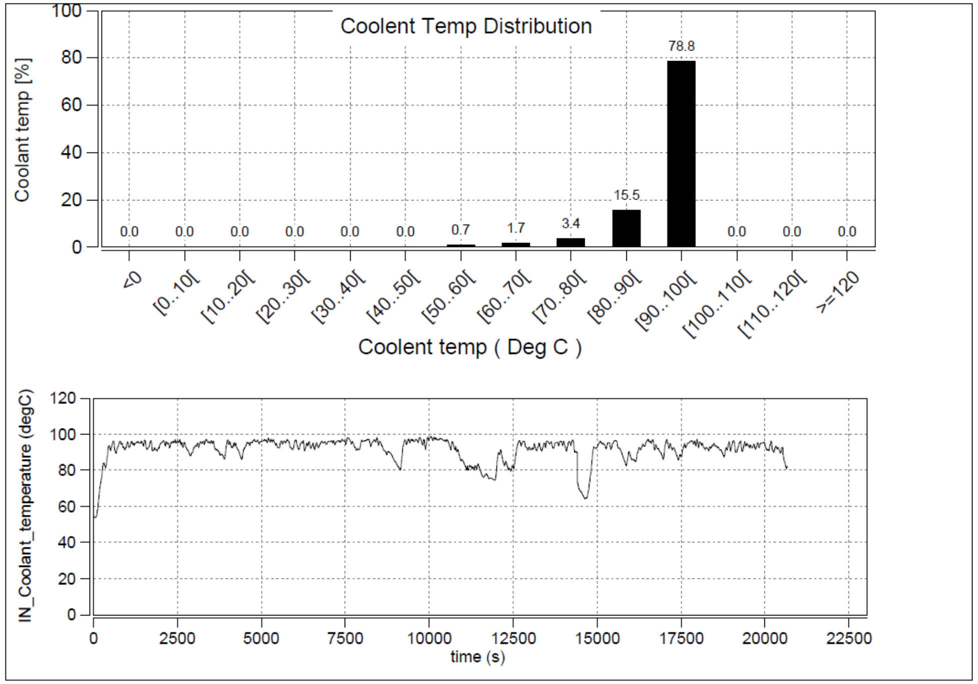
Laden Trial

Signa 4221.T RDE Test Vehicle

08.06.23

06.06.23 **55 kmph**





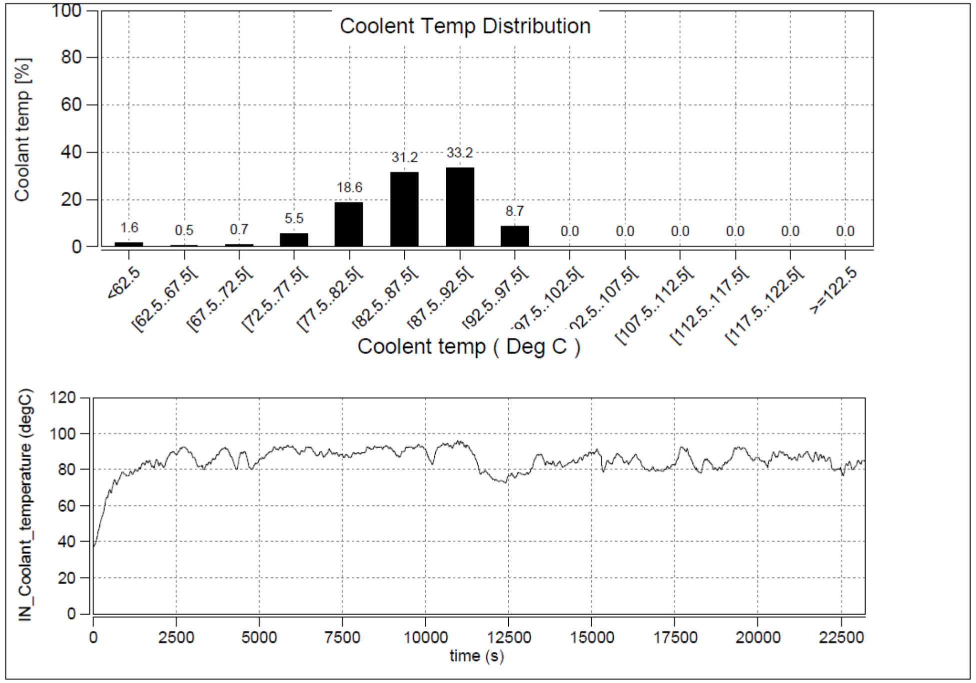
Coolant Temp Distribution

Unladen Trial

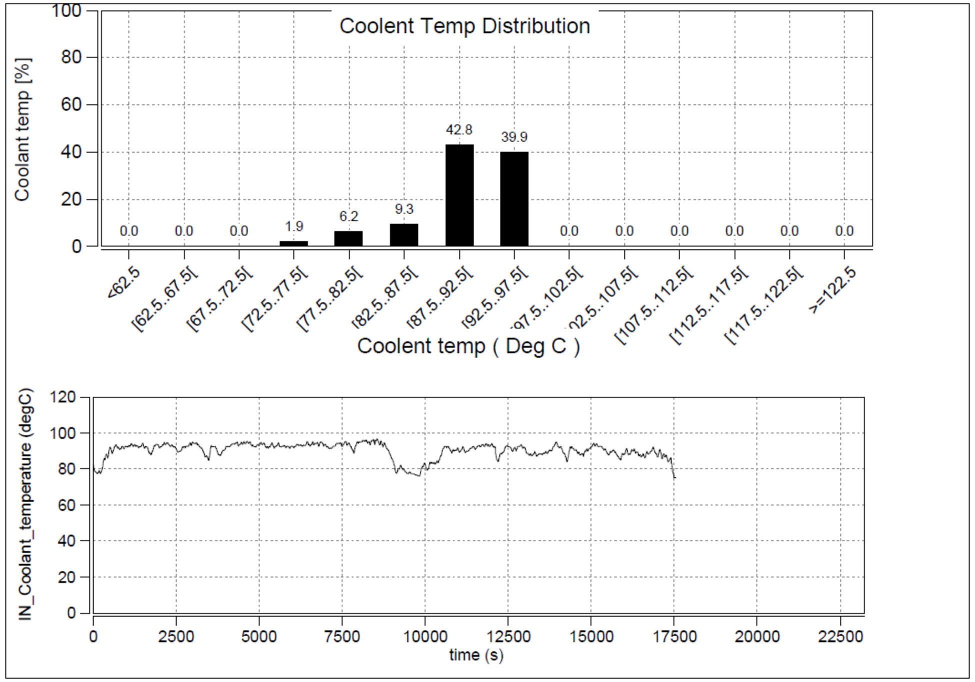
TATA MOTORS
Connecting Aspirations

Signa 4221.T RDE Test Vehicle

14.06.23	40 kmph



17.06.23	55 kmph



MORE WHEN ONE
BE BOLD | OWN IT | SOLVE TOGETHER | BE EMPATHETIC

Enclosed with FIR/Testing-Outdoor/224000/2023/6205

TATA MOTORS
Connecting Aspirations

Thank You

MORE WHEN ONE
BE BOLD | OWN IT | SOLVE TOGETHER | BE EMPATHETIC

TEST REQUISITION

Ref.	22404168R/RDE/Signa 4221.T	Date:	23/06/2023

To,

ERC Testing

Kindly arrange to take up for testing of the following component.

1	Part Description	TATA SIGNA 4221.T 5L BSVI				
2	Part Number	22404168R,				
3	Drawing number	224000000018,	Mod status	NR,1,	Mod Date	29/03/2023
4	Vehicle Model / Project	LPT 3721 5L BS4 SIGNA WITH AC RA109RR 10R20				
5	WBS Number	P.21.EA.458 -TRUCK 4425 / 4930 RDE				
6	Supplier	M/s Cummins Ind. LTD				
7	Number of samples					
8	Inspection report	Enclosed	☐ Yes	☑ No	If No commitment date:	
9	Material report	Enclosed	☐ Yes	☑ No	If No commitment date:	
10	Suppliers test report	Enclosed	☑ Yes	☐ No	If No commitment date:	
11	Reason for testing	First Source Development	☐	Weight Reduction	☐	
		Alternate source development	☐	Cost Reduction	☐	
		Data generation Test	☑	Regulatory test	☐	
		Any other (pl. specify)	☐	New Development	☐	
12	What testing required?					
13	Remarks	FE Data On Chassis dyno as per Specified duty cycle at 35-40 Cruise speed.				
14	Peripheral Parts Availability	YES				

Authorized Signature:	
Name:	
Dept:	

Contact Person:	
Phone Number	

For Testing Dept Use only	Test Item Number and Date(stamp)	Test Engineer

TATA MOTORS LIMITED ERC PUNE	TEST PLAN VEHICLE TESTING DEPARTMENT	**DATE:** 30/06/2023

Description: 22404168R/RDE/Signa 4221.T	**Test Item No:** 42355			
	New	✔	**Moderate**	x
	Major	x	**Minor**	x

Part No: 22404168R, **Drawing No:** 224000000018,	**Modification No:** NR,1,	**Enclosures:**
Make: M/s Cummins Ind. LTD	**WBS Number:** P.21.EA.458 -TRUCK 4425 / 4930 RDE	**Copies to** :
Vehicle Type: LPT 3721 5L BS4 SIGNA WITH AC RA109RR 10R20	**No. of Samples:** 1	
Test Requested by:	VEHICLE-I	**Equipments:**
Test Requisition Reference:	Testing-Outdoor/15689	
Reason for Test/s:	Data Generation	
Regulatory Requirement:	a) Component:	
	b) Vehicle:	

Sr. No.	Test description	Test specification	Acceptance criteria	No. of Samples	Test Responsibility	Proto Phase (Mule/ Alpha/ Beta/PP)	Remarks
1	Back to Back FE	As per PED trials	As per PALS	1	Debabrata Das	Beta	NA

REMARKS: Completed	**INTERNAL REF:**	
TEST TO FAILURE: No		
MEOST: No		
RELIABILITY TESTING: No		
BENCHMARKING TESTING: No		
DFMEA/FMEA DONE: Yes		
PREPARED BY	**CHECKED BY**	**APPROVED BY**